

Igor Trujnara

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 Barcelona, Spain

Employment History

Academic

- 10/2025 – ...  **Bioinformatician.** Gabaldón Lab, Institute for Research in Biomedicine, Barcelona, Spain
- Developing machine learning models for early detection of colorectal cancer using metagenomics.
- 4/2025 – 7/2025  **Master's Student.** Neutrinos Group, Institut de Física d'Altes Energies, Barcelona, Spain
- Developed deep learning models for particle classification in the HyperK experiment
 - Improved the lab's data pipeline and introduced novel training approaches
- 10/2024 – 7/2025  **Master's Student.** Guigó Lab, Centre for Genomic Regulation, Barcelona, Spain
- Explored the potential of DNA large language models for gene annotation
 - Supported other biodiversity research projects in the lab
- 1/2022 – 6/2024  **Undergraduate Researcher.** Notredame Lab, Centre for Genomic Regulation, Barcelona, Spain
- Developed a scalable and reproducible pipeline for comparing orthology annotations
 - Analyzed experimental and generated structures of the BicD2 protein, leading to novel wet lab experiments
 - Provided technical assistance to multiple projects in the group

Other

- 1/2022 – ...  **Lead Artist – Track Racer.** Vistula Development
- One of the 2 developers of an indie video game
 - Responsible for all user interfaces and visual effects

Community Roles

2024 – ...  **Maintainer.** nf-core

- Maintainer in a large open-source community (over 1k contributors and 10k users)
- Regularly providing code review to contributors
- Involved in strategic decisions on the development of the community

Education

- 9/2024 – 7/2025  **MSc Multidisciplinary Research in Experimental Sciences**, Barcelona Institute of Science and Technology
Thesis title: *Exploring the detection of DNA features by the Evo large language model*
Minor project: *Particle classification with ND280 HA-TPC using deep learning*
- 9/2021 – 7/2024  **BSc Bioinformatics**, Pompeu Fabra University
Thesis title: *nf-core/reprotho: a pipeline for comparative analysis of ortholog predictions.*

Research Publications

Journal Articles

-  L. Santus, J. Espinosa-Carrasco, L. Rauschnig, *et al.*, “An nf-core framework for the systematic comparison of alternative modeling tools: The multiple sequence alignment case study,” *NAR Genomics and Bioinformatics*, vol. 7, no. 3, lqaf104, Jul. 2025, ISSN: 2631-9268.  DOI: 10.1093/nargab/lqaf104. eprint: <https://academic.oup.com/nargab/article-pdf/7/3/lqaf104/63899826/lqaf104.pdf>.

Conference Proceedings

-  I. Trujnara, L. Santus, J. Espinosa-Carrasco, A. Vignoli, and C. Notredame, “Nf-core/reprotho: A pipeline for comparative analysis of ortholog predictions [conference poster],” en, Publisher: Zenodo, Oct. 2024.  DOI: 10.5281/ZENODO.15078564. (visited on 03/24/2025).

Skills

Natural Languages	 NATIVE: Polish FLUENT: English, Spanish, Catalan, French CONVERSATIONAL: German BASIC: Italian, Russian
Programming Languages	 ADVANCED: Python, C++, Nextflow INTERMEDIATE: R, JavaScript, Bash, $\LaTeX$ BASIC: C, Haskell, C#
Frameworks	 PYTHON: Pytorch, Pandas, Polars, Scikit, Numpy, Biopython R: Tidyverse, Shiny, Bioconductor C++: Unreal Engine JAVASCRIPT: Next.js, React.js, Tailwind
Operating Systems	 Windows, Linux
Software	 Visual Studio, VS Code, R Studio, Git, GitHub, Docker
Bioinformatics	 Workflow management, data mining, statistical models, machine learning, MSA, metagenomics, high performance computing

Skills (continued)

Soft skills  Teamwork, time management, organization, creativity, quick learning, information retrieval, work with documentation

Miscellaneous

Awards and Achievements

2024  **BIST Fellowship.** Barcelona Institute of Science and Technology, covers full tuition for the master's degree

 **BDBI Final Project Award.** Pompeu Fabra University, awarded to the best final projects among bioinformatics students

References

Prof. Cedric Notredame

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Prof. Hafid Laayouni

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Prof. Roderic Guigó

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